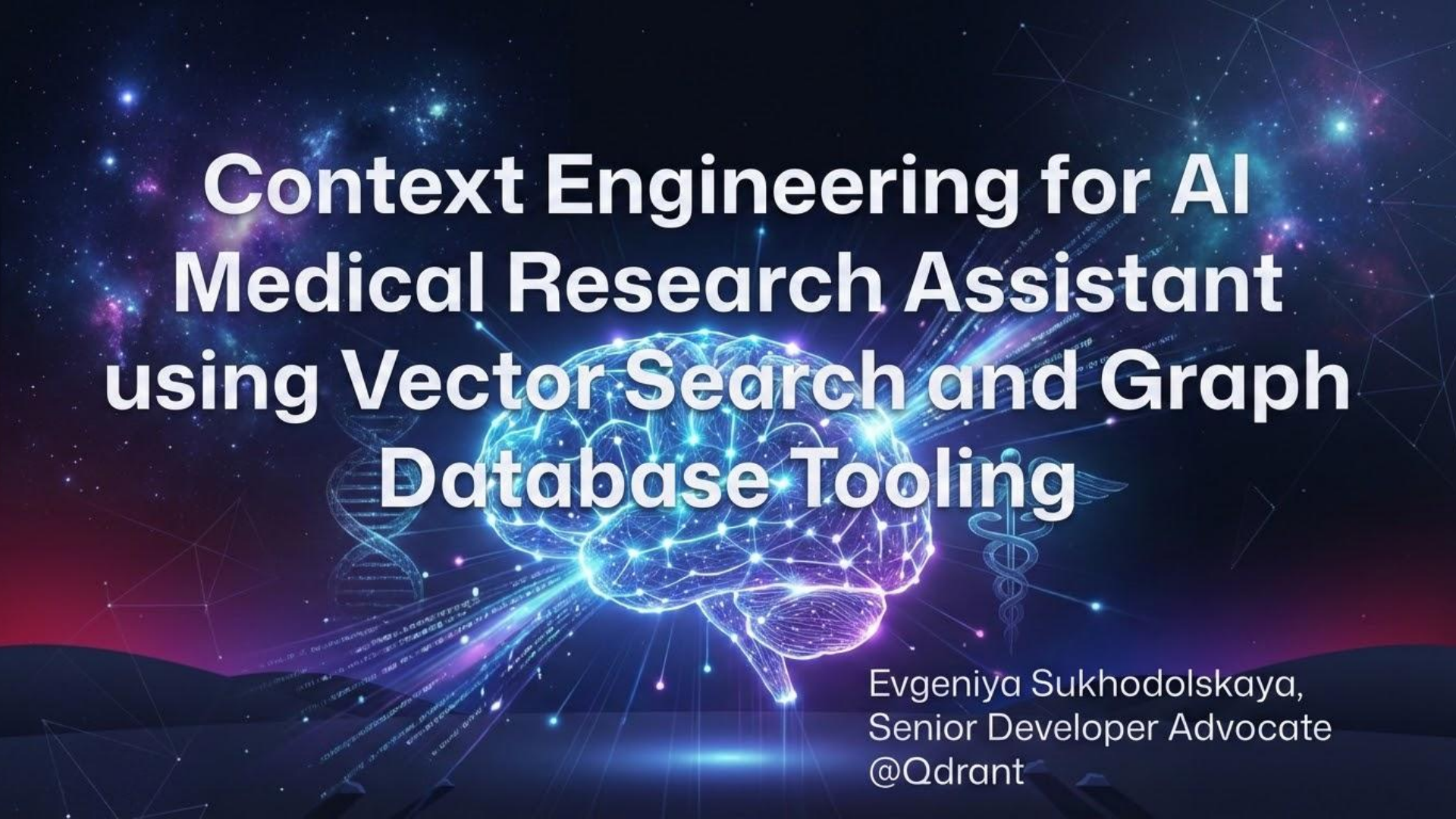


Context Engineering for AI Medical Research Assistant using Vector Search and Graph Database Tooling



Evgeniya Sukhodolskaya,
Senior Developer Advocate
@Qdrant



Neo4j for context engineering

How Graph Technology enables (Agentic) AI

Niels de Jong
PM @ Neo4j

Knowledge graphs unlock GenAI

KGs
Knowledge
Facts
Context (!)



GenAI
Language
Statistics
Creativity

Accurate
Contextual
Explainable

Neo4j: Graph Intelligence Platform

Graph AI

Aura Agents

Build multi hop agents in seconds.



Agentic Ecosystem

Integrate: AWS, GCP, Azure, LangChain, ServiceNow, Salesforce, Databricks & Snowflake



AI FRAMEWORKS



AGENT FRAMEWORKS

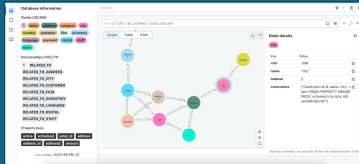
Agentic Brain

Provides foundational services like memory, context-graph to power all AI platforms.

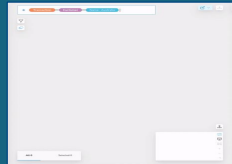


AI Powered Graph Tools

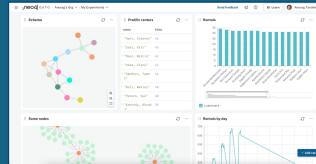
Browser (Query)



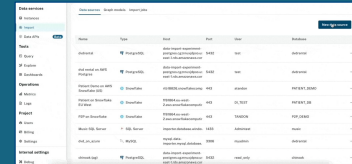
Bloom (Explore)



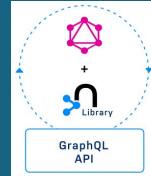
NeoDash(Dashboards)



Data & Document Import



GraphQL



Database

Intelligent Operational Database

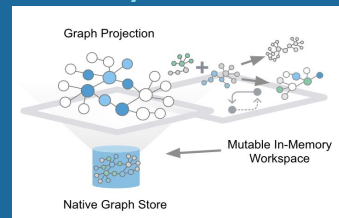
Better Predictions for Analytics



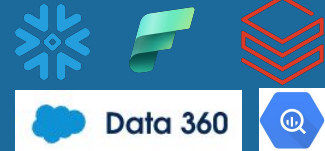
Graph Transactions, Storage & Querying

Graph Analytics, ML & Data Science

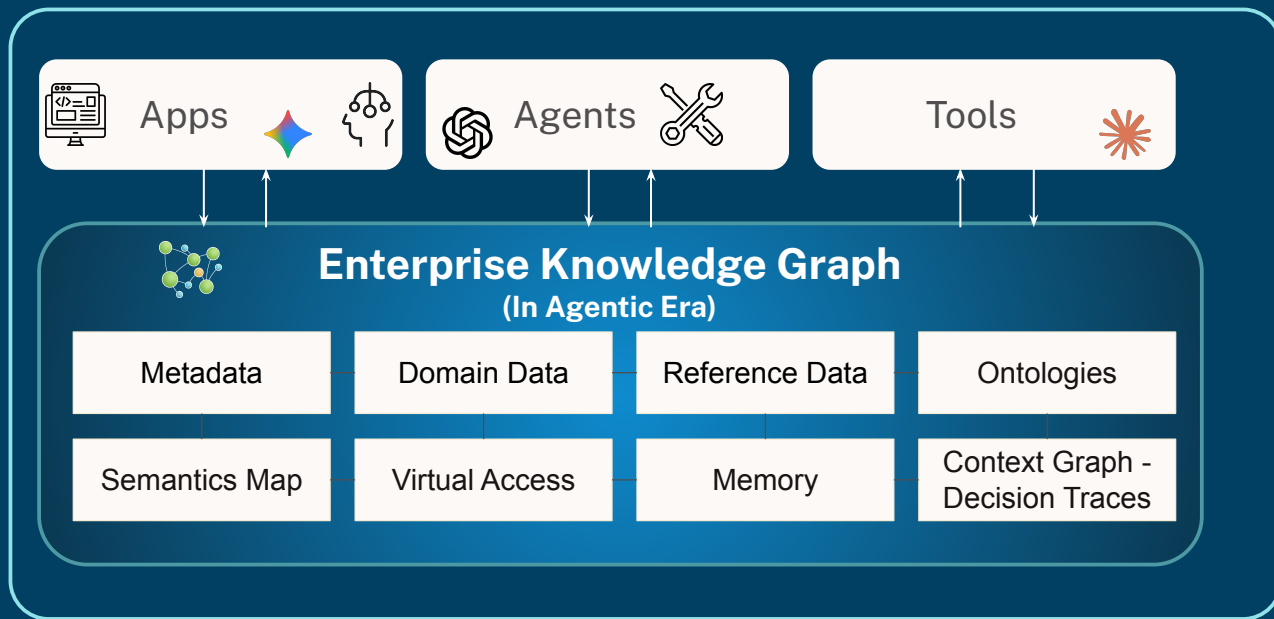
Analytics (GDS)



Graph Engine (Virtual Graph - Q2 26)

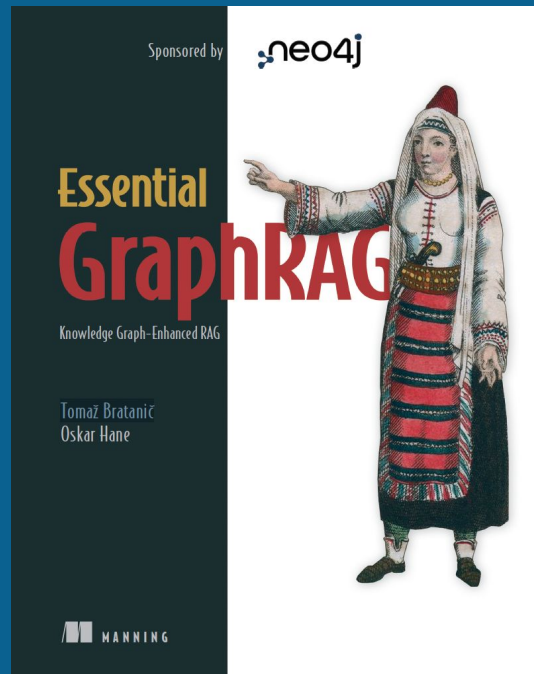
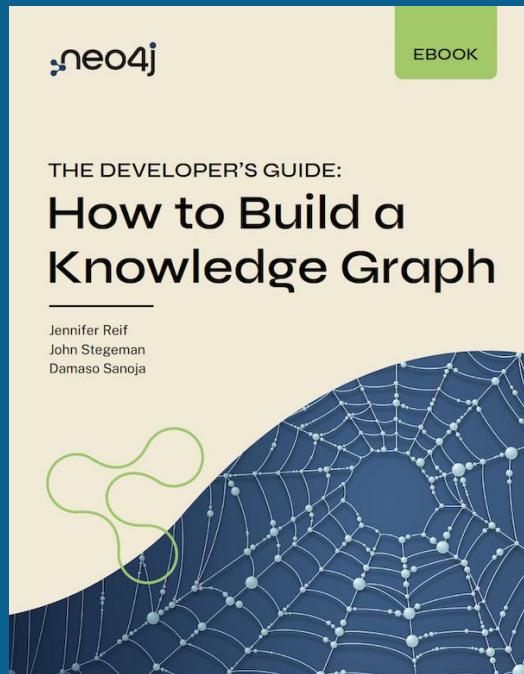


Build with Graphs



An **Enterprise Knowledge Graph (EKG)** provides the semantic layer and **relational context** necessary for agents to move from simple pattern matching to accurate, grounded reasoning.

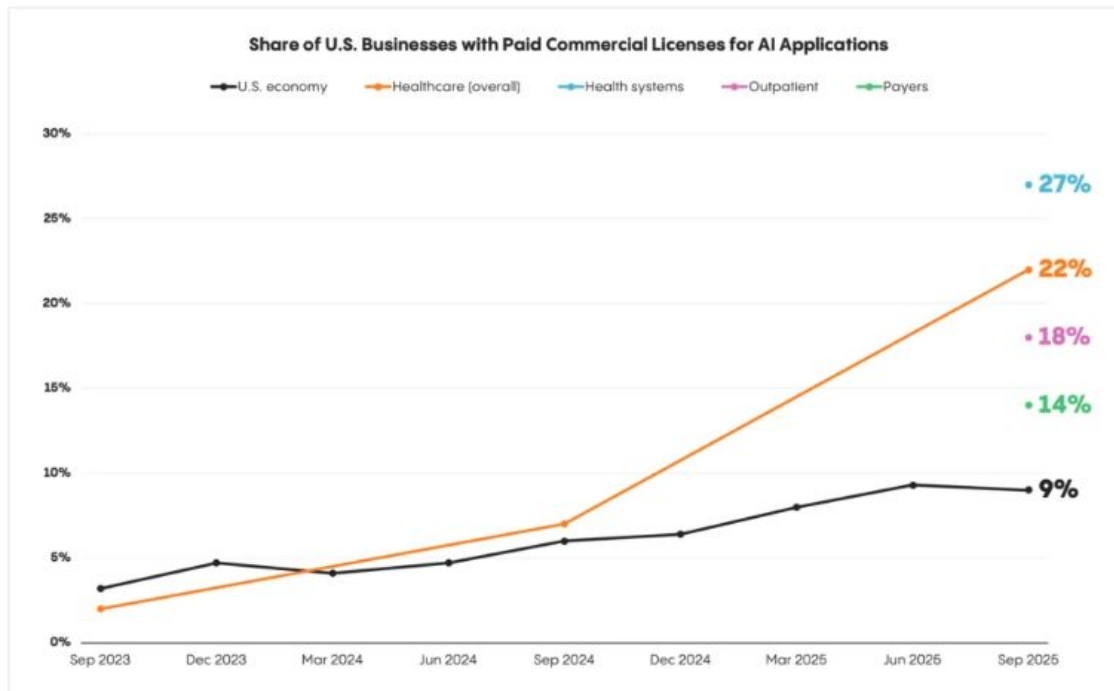
How to get started:



Why AI Medical Research Assistant?

Healthcare Is Winning the AI Race

22% of healthcare organizations have deployed commercial AI—more than double the 9% adoption rate across the U.S. economy



- Ambient scribes
- AI across the drug development lifecycle
- Medical research copilots

These solutions depend on **context mining** and **engineering**, supported by a **combination of retrieval layers**.

Use Case

Biomedical Literature Research

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

 **PubMed**[®]

Search

Advanced

User Guide

About PubMed

PubMed is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health—both globally and personally.

PubMed contains more than 39 million citations and abstracts of biomedical literature. It does not include full text journal articles; however, links to the full text are often present when available from other sources, such as the publisher's website or [PubMed Central \(PMC\)](#).

Available to the public online since 1996, PubMed was developed and is maintained by the [National Center for Biotechnology Information \(NCBI\)](#), at the [U.S. National Library of Medicine \(NLM\)](#), located at the [National Institutes of Health \(NIH\)](#).

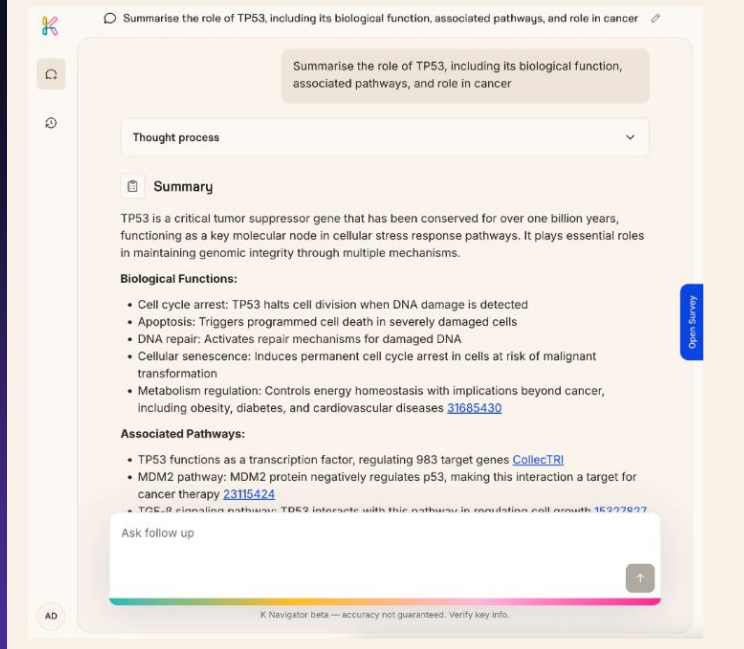
About the Content

Citations in PubMed primarily stem from the biomedicine and health fields, and related disciplines such as life sciences, behavioral sciences, chemical sciences, and bioengineering.

Biomedical Copilots

Literature Navigator – Get high quality information from scientific research

Based on millions of PubMed article abstracts



The interface shows a search bar with the query "Summarise the role of TP53, including its biological function, associated pathways, and role in cancer". Below the search bar, a dropdown menu is set to "Thought process". The main content area displays a "Summary" section with a paragraph about TP53's role as a tumor suppressor gene. Below this, there are two sections: "Biological Functions:" and "Associated Pathways:". The "Biological Functions:" section lists five bullet points: Cell cycle arrest, Apoptosis, DNA repair, Cellular senescence, and Metabolism regulation. The "Associated Pathways:" section lists two bullet points: TP53 functions as a transcription factor and the MDM2 pathway. At the bottom, there is an "Ask follow up" input field and a "K Navigator beta" footer.

Summarise the role of TP53, including its biological function, associated pathways, and role in cancer

Thought process

Summary

TP53 is a critical tumor suppressor gene that has been conserved for over one billion years, functioning as a key molecular node in cellular stress response pathways. It plays essential roles in maintaining genomic integrity through multiple mechanisms.

Biological Functions:

- Cell cycle arrest: TP53 halts cell division when DNA damage is detected
- Apoptosis: Triggers programmed cell death in severely damaged cells
- DNA repair: Activates repair mechanisms for damaged DNA
- Cellular senescence: Induces permanent cell cycle arrest in cells at risk of malignant transformation
- Metabolism regulation: Controls energy homeostasis with implications beyond cancer, including obesity, diabetes, and cardiovascular diseases [31685430](#)

Associated Pathways:

- TP53 functions as a transcription factor, regulating 983 target genes [CollecTRI](#)
- MDM2 pathway: MDM2 protein negatively regulates p53, making this interaction a target for cancer therapy [23115424](#)
- TP53 also interacts with the MDM2 pathway to regulate cell growth [16927897](#)

Ask follow up

K Navigator beta — accuracy not guaranteed. Verify key info.

AI Agents for Scientific Discovery

Crow - Concise Search

Request Limit Increase

In the Ingraham 2022 protein design paper, how did they force the protein complexes to be symmetric with respect to a specific space group

Food for thought:

> What are some likely mechanisms by which mutations near the HTRA1 locus in humans might be causal for age-related macular degeneration?

> How might you capture electron transfer effects using classical force fields for molecular dynamics simulations of protein-protein interactions?

> How compelling is genetic evidence for targeting PTH1R in small cell lung cancer?

> What factors limit the wavelengths of light detectable by mammalian eyes?

Biomedical CoPilot

GraphRAG search over biomedical literature

Biomedical CoPilot • 19:13:55

Key Findings

- 1 The **TP53** gene is mutated in approximately 50% of human cancers, and its role extends beyond tumor suppression to include regulation of apoptosis, cell cycle arrest, and DNA repair. Notably, studies show that mice lacking critical effectors of p53-induced apoptosis do not develop tumors, suggesting alternative mechanisms of tumor suppression [PMID:29149101](#) 🔗 .
- 2 Recent advancements in pharmacological strategies to activate **p53** have shown promise in treating malignancies, particularly those with **TP53** mutations. Approaches include MDM2 degraders and gene therapies aimed at restoring **p53** function, with several agents progressing to clinical trials [PMID:40602221](#) 🔗 .
- 3 Mutant **p53** (mutp53) not only loses its tumor suppressor function but also gains oncogenic properties, contributing to genetic instability and poor prognosis in cancers. This highlights the need for targeted therapies that can effectively address the challenges posed by mutp53 in clinical settings [PMID:36400749](#) 🔗 .

🔗 Graph Insights

- The knowledge graph revealed that **TP53** is frequently mentioned in 12 shared papers, indicating a robust network of research surrounding this gene.
- Collaborators such as **Marco J Herold**, **Gemma L Kelly**, **Ana Janic**, and **Andreas Strasser** are actively publishing on topics related to **TP53**, suggesting a collaborative effort in understanding its role in cancer.

💡 Synthesis

The retrieved literature underscores the critical role of **TP53** in cancer progression, highlighting its dual function as a tumor suppressor and an oncogenic factor when mutated. The knowledge graph enriches this understanding by illustrating the collaborative research efforts focused on **TP53**, as well as its extensive involvement in various studies. This interconnectedness emphasizes the importance of ongoing research into therapeutic strategies targeting both wild-type and mutant **p53**. However, gaps remain in fully elucidating the mechanisms by which mutp53 contributes to tumorigenesis and how best to counteract its effects in clinical settings.

4 tools executed 3 Matches

Ask about genes, diseases...

📌 TOP K 1 3 5

Vector Search Results Trace Hide

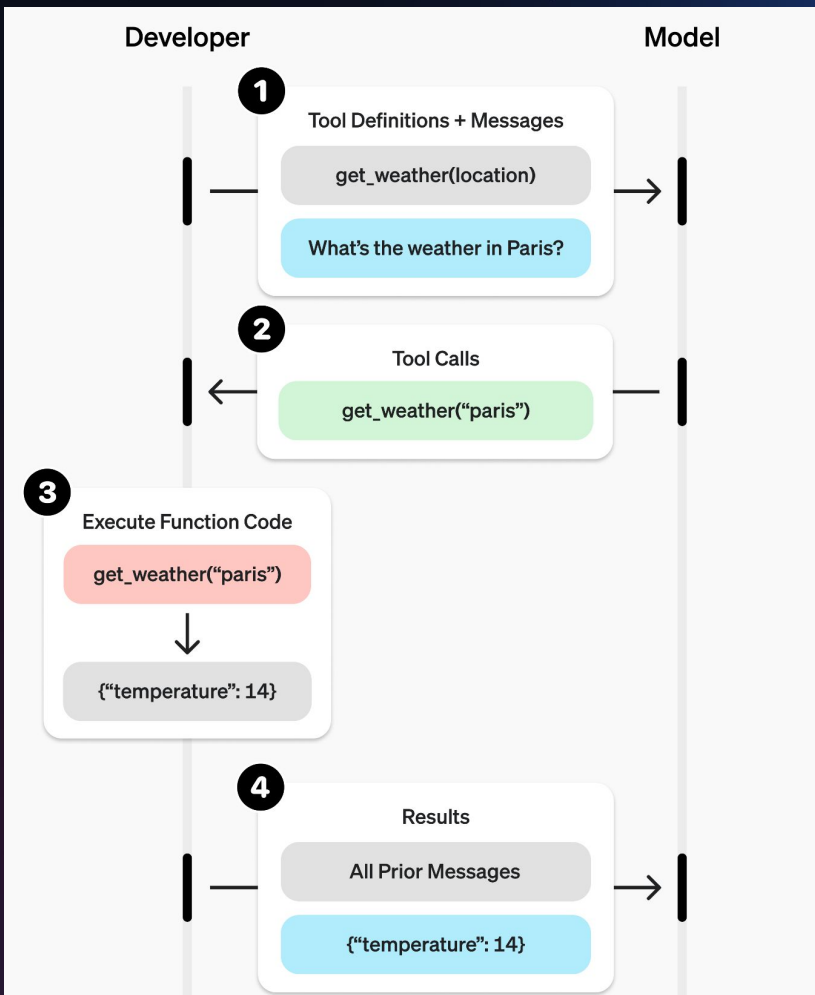
🛠️ 4 tools executed

- > 🔵 retrieve_papers_hybrid 3 Matches
- > 🔵 get_genes_in_same_papers 10 Matches
- > 🔵 get_collaborators_with_topics 4 Matches
- > 🔴 summarize

Agenda

- Which **Qdrant** and **Neo4j** tools are implemented now in **PubMed Navigator** and why
- **Demo:** How LLM uses these tools for context mining
- **Hands on: Qdrant**-related choices for context engineering in systems like **PubMed Navigator**.
 - Ensuring Scalability & Results Relevance
 - Tips on Tools & Prompts Engineering

“Tools are a new kind of software which reflects a contract between deterministic systems and non-deterministic agents.”



Query 1: *“Explain the role of TP53 in cancer progression”*

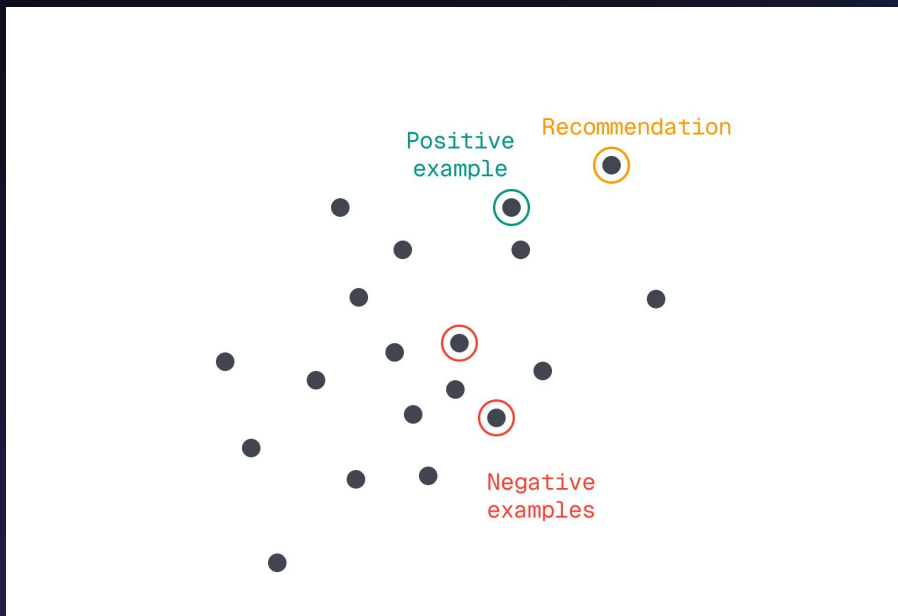
Query 2: *“Explain the role of TP53 in cancer progression, **but excluding animal studies**”*

Query 3: *“**Which authors have collaborated on studies** of TP53 mutations in adenoid cystic carcinoma”*

Need in Vector Search

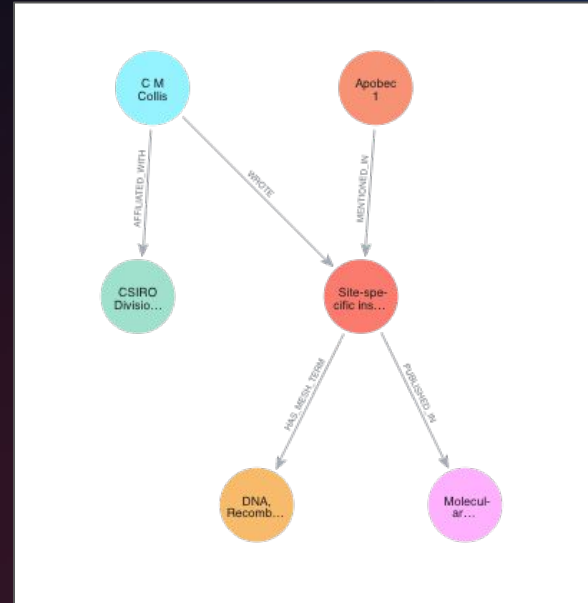
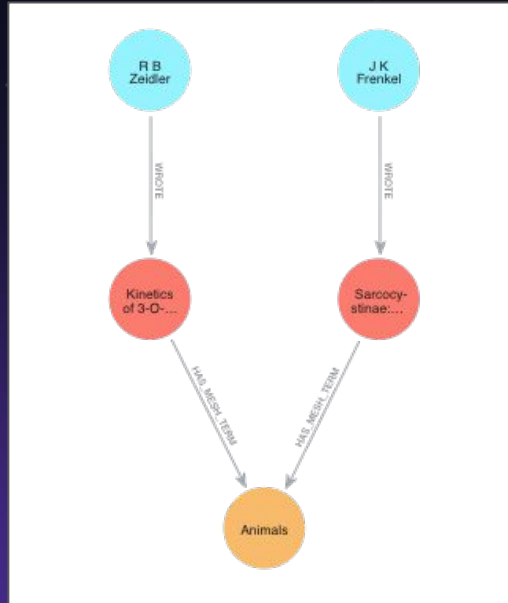
Query 1: “Explain the role of TP53 in cancer progression.”

Query 2: “Explain the role of TP53 in cancer progression,
but excluding animal studies.”

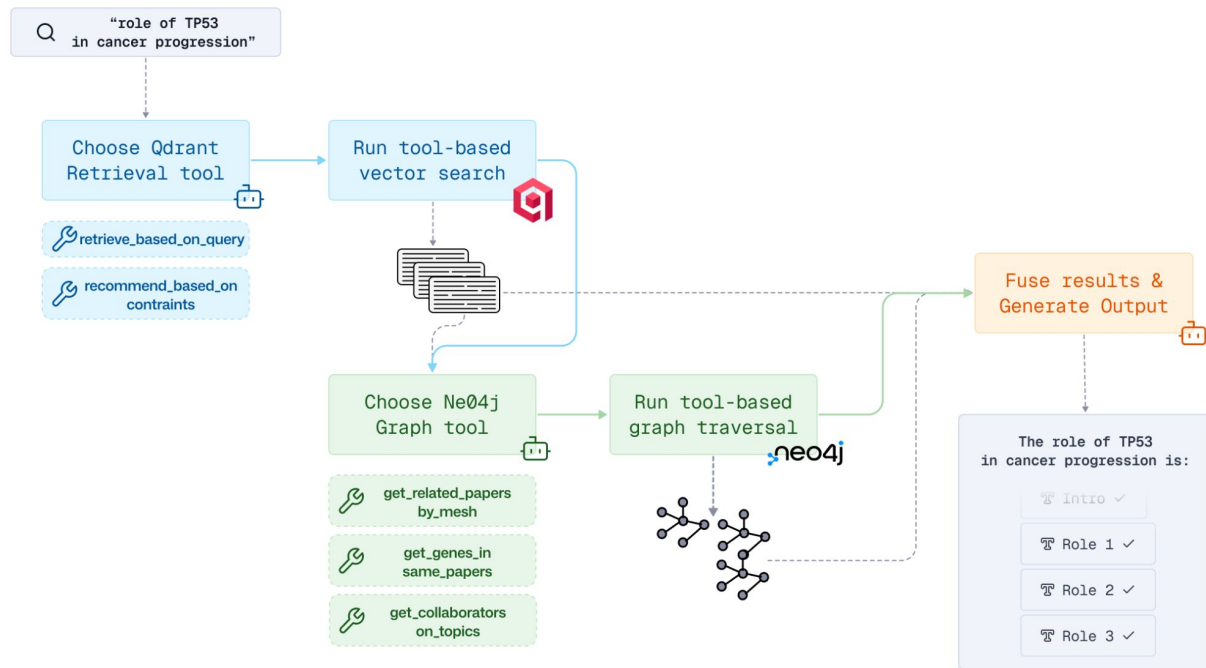


Need in Knowledge Graphs

Query 3: “Which authors have collaborated on studies of TP53 mutations in adenoid cystic carcinoma”



PubMed Navigator





Demo

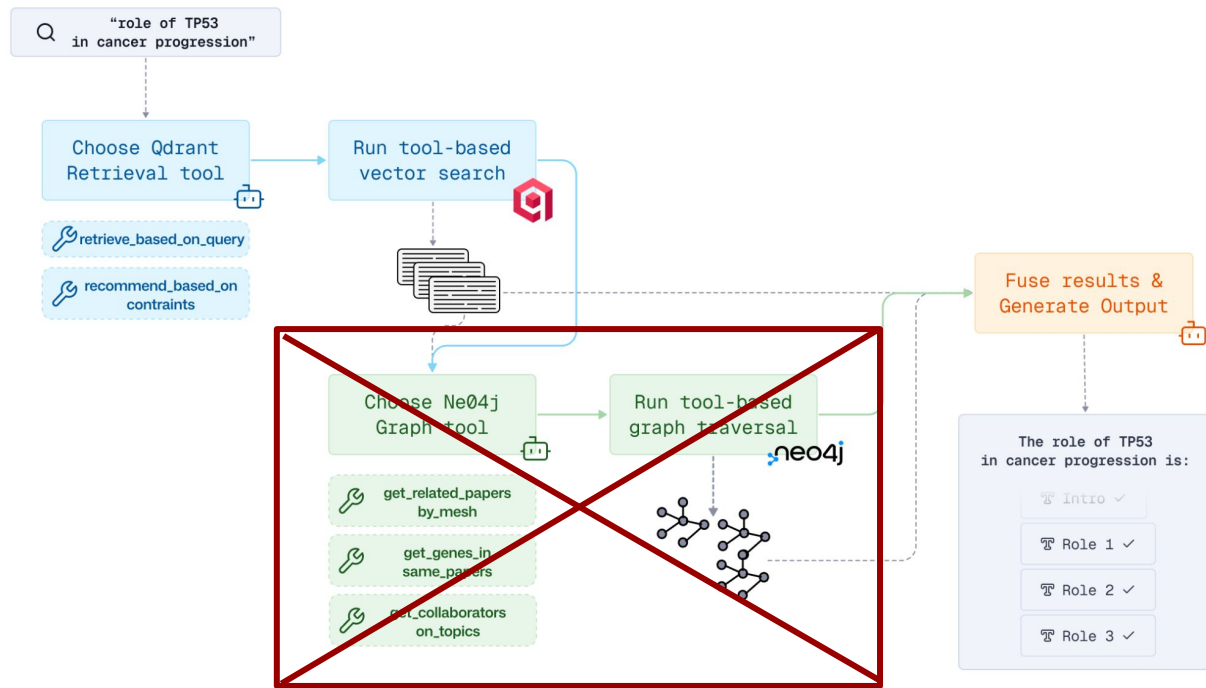
*(Praying demo
gods...)*



Hands On:

Qrant-related Choices for Context Engineering in Systems like PubMed Navigator.

What We Will Build



What We Will Learn

Hands on: Qdrant-related choices for context engineering in systems like **PubMed Navigator**.

- Ensuring Vector Search Scalability & Results Relevance
- Tips on Tools & Prompts Engineering

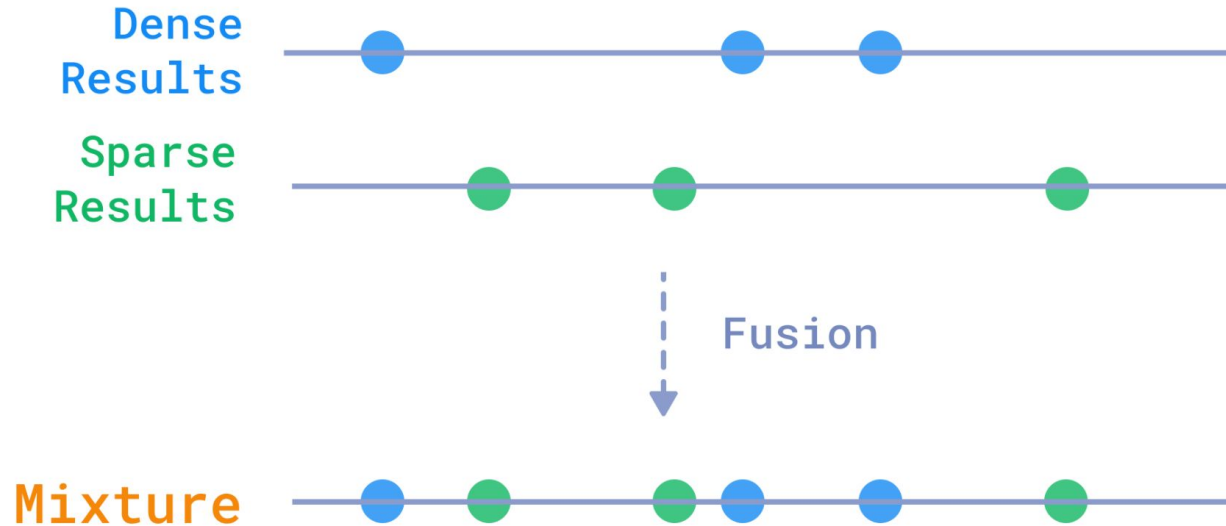


What We Will Use

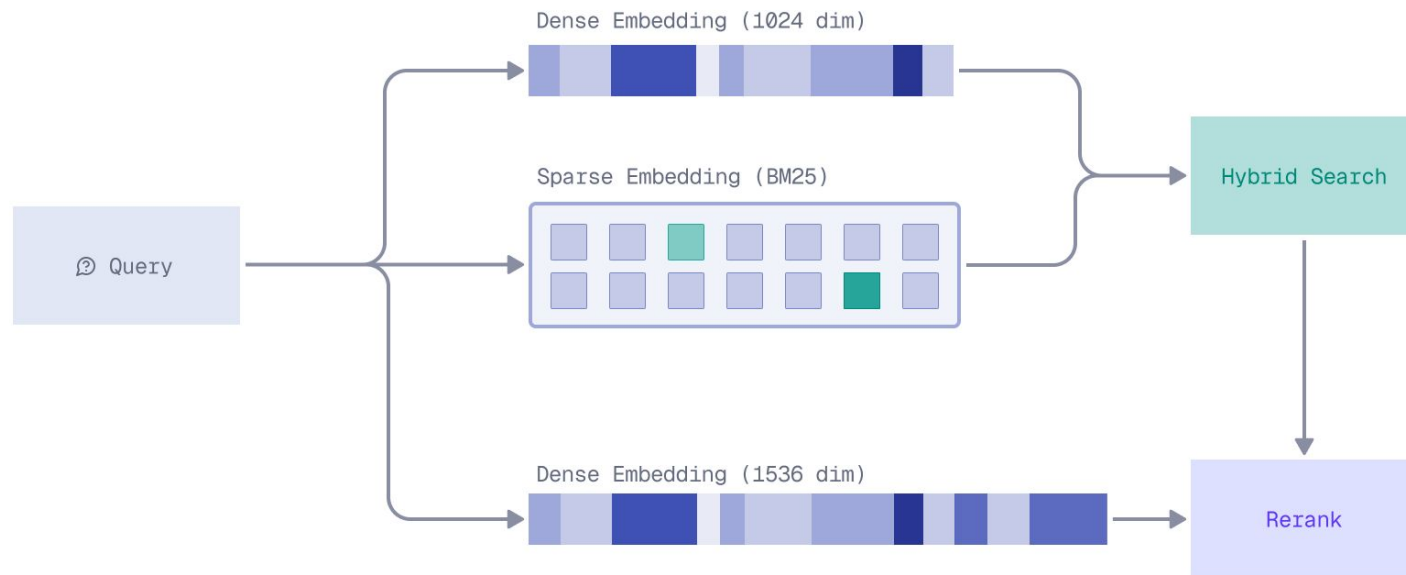
- Python 3.13+
- uv package manager
- 27k PubMed Papers dataset (163 MB uncompressed)
- OpenAI models (embeddings & LLM) (~**0.17** cents for the end-to-end workshop run)
- Qdrant Managed Cloud (Free Tier);



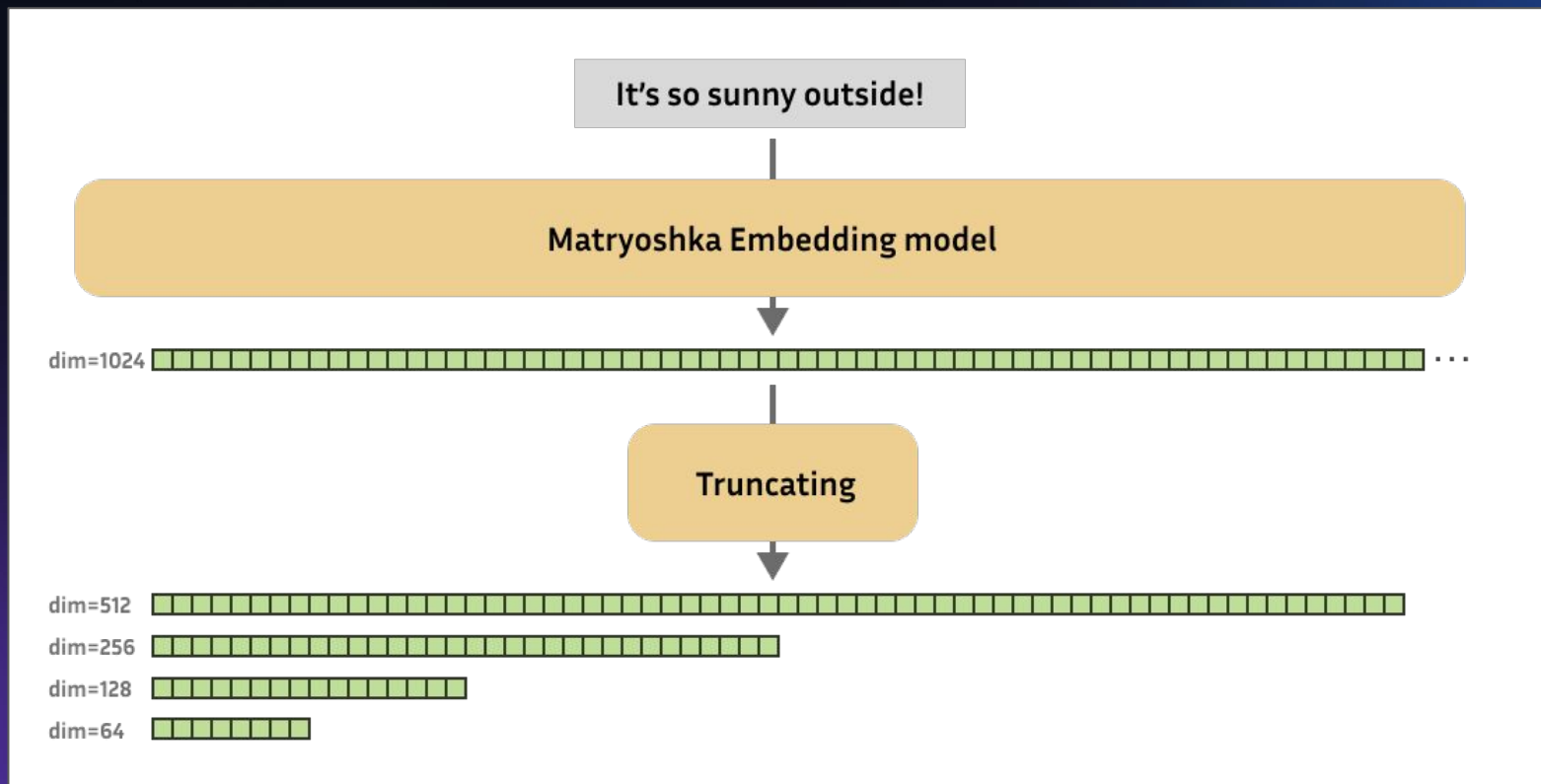
Hybrid Search: Keywords and Semantics



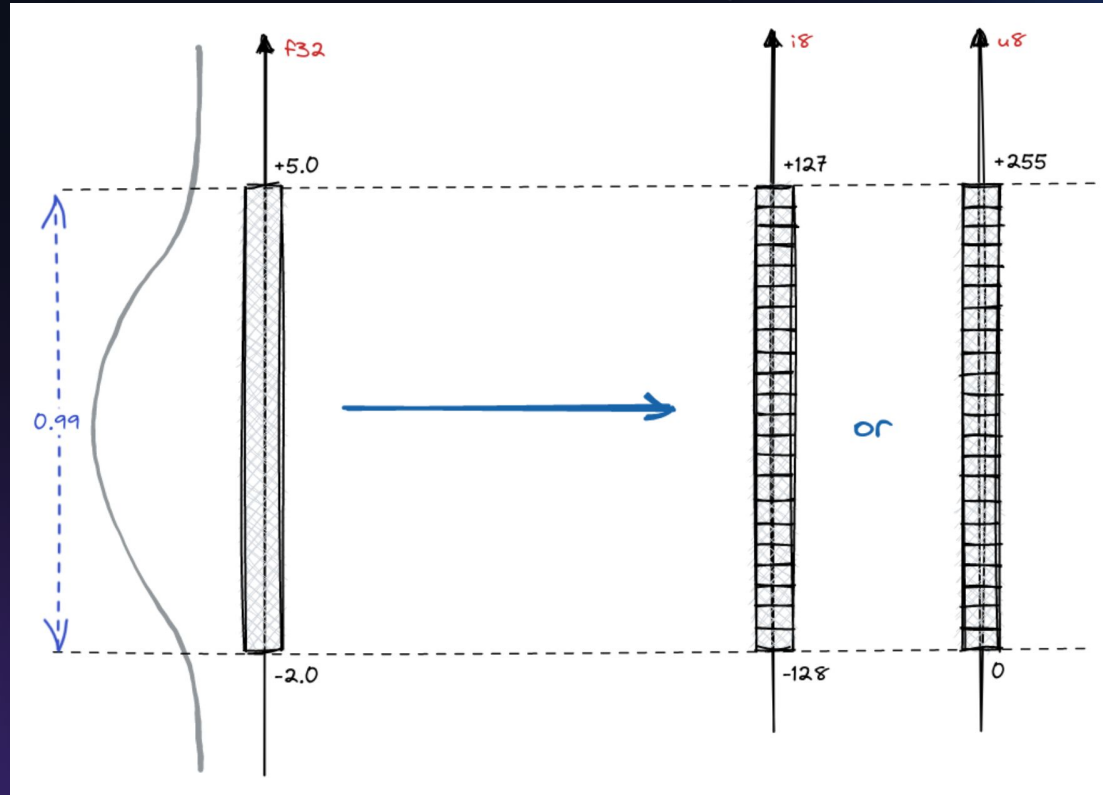
Hybrid Search Fused by Reranking



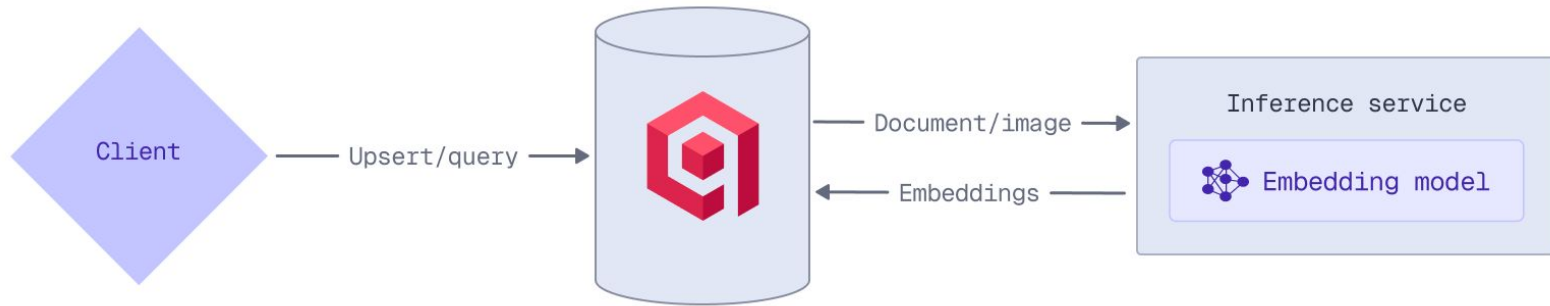
Matryoshka Representation Learning (MRL)



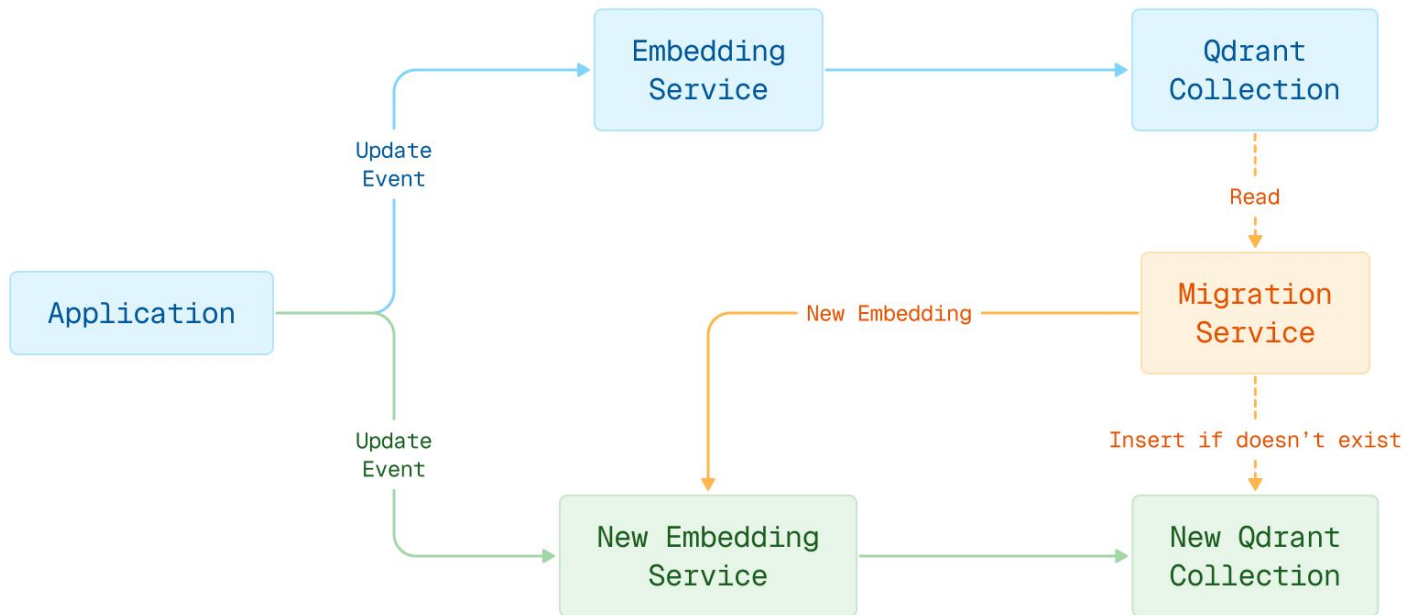
Scalar Quantization



Cloud Inference



Conditional Updates/Uploads



To Sum Up Tips & Tricks

Prompt Engineering Best Practices

- Think of an LLM as a good new hire. It needs context!
 - Explain the **why** behind instructions (not just screaming in caps lock)
 - Optimize prompts with (other) LLMs by asking to ask you questions
- Clear structure: Identity → Instructions → Examples → Context
 - XML tags/markdown
 - 3-5 examples: relevant and diverse
 - Divide general instructions/context (“developer”/“user”)
 - Long data goes at the top, mind lost-in-the-middle
- Your prompt writing style affects the answer style

Tool Engineering Best Practices

- Each tool should have a clear, distinct purpose, functionality shouldn't overlap!
 - Think of real examples. If you don't know which tool you'd choose, LLM won't either.
- Offload the burden from LLMs to code.
 - If you know how an argument should be filled, populate it directly in code.
- Prompt engineer tools descriptions:
 - 3-4 sentences minimum
 - When to use and when NOT to use (here scream in caps).
 - Tool naming actually matters!
- Let agents (Claude <3) analyze and improve your tools

Vector Search Scalability & Quality (Workshop Features)



- **Search Quality:**
 - **Hybrid search** (semantic + lexical)
 - **Multistage retrieval** with reranking using **Matryoshka Representation Learning (MRL)**
- **Search Scalability:**
 - **Scalar quantization** with oversampling and rescoreing
- **Efficient data uploading:**
 - **Qdrant Cloud Inference** (with OpenAI's MRL)
 - **Conditional uploads/updates**
 - **Lazy batching** with **parallelism & retries**

and **Recommendation API** to see vector search at its full potential.



Evgeniya Sukhodolskaya
Dev Advocate at Qdrant



Thank You!

